L2.1 Maximum and Minimum Values

Work in your group. Check your answers against the Quarto tonight.

Local and absolute extrema

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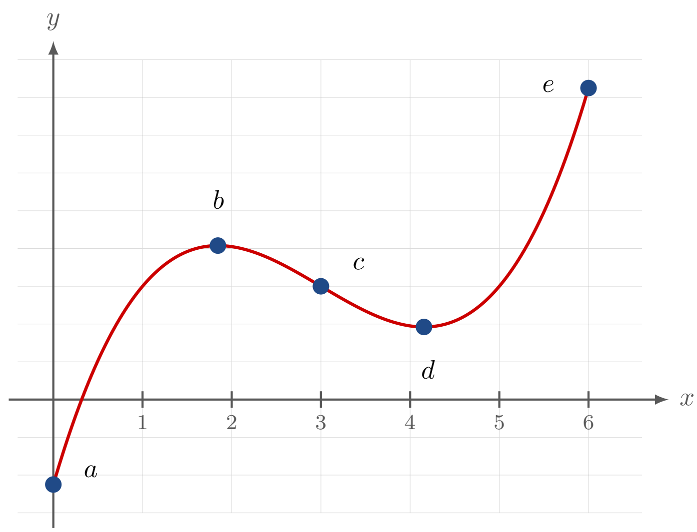
1. For each marked point, give **every** label that applies — local max, local min, absolute max, absolute min, or neither.



2. Sketch the graph of a function that has two local maxima, one local minimum, and **no absolute minimum**.

3. Sketch, by hand, and read off every absolute and local extremum.

$$f(x) = \begin{cases} 2x+1, & 0 \leqslant x < 1 \\ 4-2x, & 1 \leqslant x \leqslant 3 \end{cases}$$

Stop. At $x = 1$, check the values just to the *left* and just to the *right* separately before you name it.

4. On the interval $[-1, 2]$, sketch a function that has

(a) an absolute maximum but **no** local maximum

(b) a local maximum but **no** absolute maximum

The Extreme Value Theorem

5. Does the Extreme Value Theorem apply? If it does not, say **which hypothesis fails**.

(a) $f(x) = x^2 - 4x$ on $[0, 5]$

(b) $f(x) = \dfrac{1}{x-2}$ on $[0, 5]$

(c) $f(x) = x^3$ on $(0, 4)$

(d) $f(x) = |x|$ on $[-2, 2]$

6. Sketch $f(x) = \dfrac{1}{x}$ on the open interval $1 < x < 3$ and read off its absolute and local extrema. Then say which hypothesis of the Extreme Value Theorem this function fails, and **which extremum that costs it**.

Fermat's Theorem

7. Find the critical numbers.

(a) $f(x) = 4 + \frac{1}{3}x - \frac{1}{2}x^2$

(b) $f(x) = x^3 + 6x^2 - 15x$

(c) $f(x) = 2x^3 + x^2 + 2x$

Stop. A critical number is not only where $f' = 0$. What is the other half of the definition?

8. Find the critical numbers. Both of these have one that the other half of the definition catches.

(a) $h(t) = t^{3/4} - 2t^{1/4}$

(b) $F(x) = x^{4/5}(x-4)^2$

9. $f(x) = 5 - (x-3)^2$ on $[0, 6]$. **Prove** that f has a horizontal tangent at $x = 3$ — do not just observe it. Name the theorem, check each of its hypotheses, then use its conclusion.

The Closed Interval Method

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10. Find the absolute maximum and absolute minimum values of $f(x) = x^3 - 6x^2 + 5$ on $[-3, 5]$.

11. Find the absolute maximum and absolute minimum values of $f(t) = t - \sqrt[3]{t}$ on $[-1, 4]$.

> **Stop.** Where is f' zero, and where does f' fail to exist? Both kinds go on the candidate list.

Lawyer Math

12. BOARDS **Group notes.** For each theorem below: write the statement **exactly**, list its hypotheses one per line, and then do the problem beside it.

② **Extreme Value Theorem.** Suppose f is continuous on a closed interval $[a, b]$. (a) Which theorem guarantees that an absolute maximum value and an absolute minimum value exist? (b) What steps would you take to find them?

③ **Fermat's Theorem.** Find the critical numbers of $h(p) = \dfrac{p-1}{p^2+4}$.

④ **Closed Interval Method.** Find the absolute maximum and minimum values of $f(x) = 5 + 54x - 2x^3$ on $[0, 4]$.

① **IVT** is already in your notes. ⑤ **Rolle** and ⑥ **MVT** are the next two.